House Price Prediction System

MODEL DESIGN DOCUMENT (MDD)

1.1 Title of the Project:

HOUSE PRICE PREDICTION USING MACHINE LEARNING

1.2 Problem Statement:

The objective of this project is to develop a machine learning-based system for predicting house prices using various property-related features such as square footage, number of bedrooms (BHK), property condition, and location. In the real estate sector, accurate price estimation is very important for buyers, sellers, and investors to make informed decisions.

Traditional methods of estimating house prices are often based on manual evaluation, personal experience, or market assumptions, which can be time-consuming and may not always be consistent or reliable. These approaches may fail to capture the complex relationships between multiple factors that influence property prices.

To overcome these limitations, this project applies machine learning techniques to automatically learn patterns from data and generate more accurate and consistent predictions. Multiple regression models are implemented and compared to evaluate their performance.

The final system is integrated into an interactive web-based application developed using Streamlit, where users can input property details and obtain real-time price predictions. This makes the solution practical, user-friendly, and accessible for real-world use.

1.3 Objectives:

- To develop a machine learning model capable of predicting house prices based on important property features such as area, number of bedrooms (BHK), property condition, and location.
- To analyse and understand how different input features influence house prices, and identify which features contribute most to the prediction.
- To implement and compare multiple regression models (such as Linear Regression and Random Forest) in order to evaluate their performance using suitable metrics like RMSE and R^2 score.
- To select an appropriate model by considering both prediction accuracy and practical constraints such as model size and deployment feasibility.
- To design and develop an interactive web-based application using Streamlit that allows users to input property details and obtain real-time price predictions.
- To enhance the user interface by incorporating dashboard elements such as sliders, column layouts, and basic visualisations for a better user experience.
- To deploy the final application on a cloud platform so that it can be accessed online without requiring local setup.

2. Brief on the project:

This project is a machine learning-based regression problem that focuses on predicting house prices using various property-related features. The primary problem addressed in this project is the difficulty in accurately estimating property prices due to the influence of multiple factors such as area, number of rooms, construction status, and location. Accurate price prediction is important for both buyers and sellers in the real estate market, as it helps in making informed decisions and avoiding overpricing or underpricing of properties.

The motivation behind this project lies in the growing demand for data-driven solutions in the real estate sector. With increasing urbanisation and property investments, having a reliable predictive model can significantly improve decision-making. Several previous studies and applications have used machine learning techniques such as Linear Regression and ensemble methods for price prediction, showing promising results in capturing relationships between features and house prices.

In this project, a structured approach is followed, starting from data collection and preprocessing, followed by exploratory data analysis to understand patterns and relationships. Machine learning models such as Linear Regression and Random Forest are implemented and evaluated using performance metrics like RMSE and R^2 score. Finally, the best-performing model is deployed using Streamlit to create an interactive web application that allows users to input property details and obtain real-time price predictions.

3. Deliverables of the project:

3.1 Approach to solve the problem:

In this project, a systematic approach is followed to solve the problem of house price prediction using machine learning. The first step involves understanding the dataset and identifying the important features that affect house prices. After that, data preprocessing is performed where unnecessary columns are removed, and categorical values are converted into numerical form so that they can be used by machine learning algorithms. Boolean values are also converted into 0 and 1 for consistency.

Once the data is cleaned, exploratory data analysis (EDA) is carried out. In this step, different types of graphs, such as distribution plots, scatter plots, and correlation heatmaps are used to understand how the data behaves. For example, the distribution plot helps in identifying whether the price data is skewed, while the scatter plot helps in understanding the relationship between square footage and price. The heatmap shows how strongly different features are related to each other.

After understanding the data, machine learning models are built. In this project, two models are used: Linear Regression and Random Forest Regressor. The dataset is divided into training and testing sets using a train-test split. The models are trained on the training data and then tested on unseen data to evaluate their performance.

The evaluation of models is done using metrics such as RMSE (Root Mean Squared Error) and R^2 Score. RMSE gives the average prediction error, while R^2 shows how well the model explains the variation in data. These metrics help compare the models and select the best one.

Based on evaluation results and practical constraints, Linear Regression is selected for deployment due to its smaller size and efficiency. A simple web application is created where users can input property details like area, BHK, and location, and the model predicts the house price in real time. The overall effectiveness of the solution is shown through model accuracy, evaluation metrics, and working application.

3.1 List of Questions Addressed:

The project is designed to answer the following important questions:

- How does the size of the house (square feet) influence its price?
- Which features play the most important role in determining house prices?

- Can machine learning models be used to predict house prices accurately?
- Which model gives better performance for this dataset: Linear Regression or Random Forest?
- How well does the model perform when tested on new unseen data?

3.2 Model Details, Findings, and Expected Outcome:

In this project, two machine learning models are implemented: Linear Regression and Random Forest Regressor. Linear Regression is used as a basic model to understand the linear relationship between input features and house price. However, real-world data is often complex and not perfectly linear.

Random Forest is a more advanced model that uses multiple decision trees and combines their results. This helps in capturing non-linear relationships and improves prediction accuracy. After training and testing both models, it is observed that Random Forest performs significantly better than Linear Regression.

The evaluation results show that Random Forest has a lower RMSE and a higher R^2 score, indicating more accurate predictions. One of the important findings from the analysis is that square footage has a strong impact on house price. It is also observed that the price data is skewed due to the presence of very high-value properties.

During the development and analysis of this project, several important observations are made:

- The area of the property (square feet) has a strong influence on the house price.
- The number of bedrooms (BHK) also significantly affects pricing.
- Location plays an important role, as properties in different coordinates show variation in prices.
- Random Forest model performs better in terms of accuracy but is not practical for deployment.
- Linear Regression provides a good balance between performance and efficiency, making it suitable for real-world applications.

The expected outcome of this project is a working system that can predict house prices based on user inputs. The application should allow users to enter details such as area, BHK, and property features, and instantly display the predicted price.

The system should also demonstrate how machine learning models can be applied to real-world problems and integrated into a user-friendly web interface. The addition of dashboard elements such as sliders, columns, and charts improves user interaction and makes the application more intuitive.

4. Resources:

4.1 Data source:

In this project, a housing dataset is used to build a machine learning model for predicting house prices. The dataset contains important property-related features such as square footage, number of bedrooms (BHK), construction status, resale condition, and geographical coordinates like latitude and longitude. The target variable in the dataset is the house price, which is measured in lakhs.

The dataset consists of approximately **29,451 records**, which provides a good amount of data for training and testing the model. The data represents real-world housing information, making it suitable for understanding how different factors influence property prices.

The dataset was obtained online as a CSV file for further processing and analysis.

4.2 Dataset File Name:

House Price.csv

4.3 Dataset Path:

E:\HousePriceProject\House Price.csv

4.4 Software and tools used:

Different software tools and libraries were used in this project to complete all stages of the machine learning workflow.

- **Python** was used as the main programming language because it is simple and widely used for data science and machine learning tasks.
- **Jupyter Notebook** was used to write and execute code step-by-step, which helps in better understanding and debugging.
- **Pandas and NumPy** were used for data handling, cleaning, and numerical operations.
- **Matplotlib and Seaborn** were used to create visualizations such as histograms, scatter plots, and heatmaps for data analysis.
- **Scikit-learn** was used to build machine learning models like Linear Regression and Random Forest, and also for model evaluation.
- **Streamlit** was used to deploy the trained model into a simple web application, allowing users to input data and get predictions in real time.
- **GitHub** is used to store and manage project files
- **Streamlit Community Cloud** is used to deploy the application online

These tools together helped complete the full process, from data preprocessing to model deployment.

4.5 References:

Some general references used during the project include:

- Documentation of scikit-learn for understanding regression models
- Streamlit documentation for building and deploying web applications
- Basic concepts of machine learning and regression techniques

4.6 Dataset Understanding:

The dataset contains both categorical and numerical features that describe residential properties. The target variable is TARGET(PRICE_IN_LACS), which represents the price of the property.

- **Number of Rows:** 29,451
- **Number of Columns:** 12

Feature Description:

- **POSTED_BY**: Indicates whether the property is listed by the owner or a dealer
- **UNDER_CONSTRUCTION**: Indicates if the property is under construction
- **RERA**: Shows whether the property is approved under RERA regulations
- **BHK_NO**: Number of bedrooms, hall, and kitchen
- **BHK_OR_RK**: Type of property (BHK or RK)
- **SQUARE_FT**: Total area of the property in square feet
- **READY_TO_MOVE**: Indicates whether the property is ready for occupancy

- **RESALE**: Indicates whether the property is a resale property
- **ADDRESS**: Location details of the property
- **LATITUDE & LONGITUDE**: Geographic coordinates of the property
- **TARGET(PRICE_IN_LACS)**: Price of the property in lakhs (target variable)

5. My details:

Name: Pronami Gogoi

Email ID: pronamigogoi120@gmail.com

Phone no. 8811846828

6. Exploratory data analysis:

Exploratory Data Analysis was performed to understand the dataset and identify patterns.

6.1 Price Distribution:

The distribution of house prices is right-skewed. This means that most houses fall within a lower price range, while only a few houses have very high prices.

6.2 Area vs Price:

A scatter plot was created to analyse the relationship between square footage and price. It shows a positive relationship, meaning that as the area increases, the price also increases.

6.3 Correlation Heatmap:

A heatmap was used to visualise the correlation between different features. It shows that square footage has a strong correlation with house price, while other features have a moderate influence.

7. Data Processing:

Before building the model, the data was cleaned and prepared.

- Boolean columns were converted into integer values
- Categorical variables were encoded into a numerical format
- Relevant features were selected for modelling
- Data was structured into input features and the target variable.

  The dataset was divided into independent and dependent variables. The independent variables (features) were stored in X, while the target variable (house price) was stored in y. The dataset was then split into training and testing sets using an 80:20 ratio to evaluate model performance on unseen data.

This step ensures that the dataset is suitable for machine learning algorithms.

8. Model building:

Two machine learning models were used in this project.

1. Linear Regression

Linear Regression is a simple and widely used algorithm that models the relationship between input features and the target variable using a linear equation. Although it provides lower accuracy compared to Random Forest, it has several advantages, such as simplicity, fast computation, and a very small model size. Because of these reasons, it is selected as the final model for deployment.

2. Random Forest

Random Forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy. This project, it shows strong performance with a higher R^2 score and lower prediction error. However, the major drawback of this model is its large size. After training, the model file becomes very large, which makes it difficult to upload and deploy on cloud platforms.

9. Model Evaluation:

The performance of the models was evaluated using the R^2 score and RMSE.

Results:

- Linear Regression → $R^2 = 0.40$, RMSE = 568.79
- Random Forest → $R^2 = 0.74$, RMSE = 374.02

Observation:

Random Forest performs better than Linear Regression. It explains approximately **74%** of the variation in house prices and has a lower prediction error.

10. Deployment:

The final model was deployed using Streamlit. A web application was created where users can input property details such as area, number of bedrooms, and other attributes.

The final application is uploaded to GitHub and deployed using Streamlit Cloud. This allows users to access the application online without running code locally.

🏠 House Price Prediction Dashboard

Enter Property Details Below:

Square Feet

1000

BHK

2

Under Construction

1

RERA Approved

0

Ready to Move

1

Resale

0

Longitude

77.00

Latitude

28.00

Posted By

- Dealer
- Owner

BHK Type

- BHK
- RK

🚀 Predict Price

💰 Predicted Price: 557.95 Lakhs

📊 Price Visualization